PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGION FORM ONE TERMINAL EXAMINATION
031 PHYSICS
MARKING GUIDE: - 2026.

1:

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	B	A	D	D	C	B	D	C	B

01@ = 10mark

2:

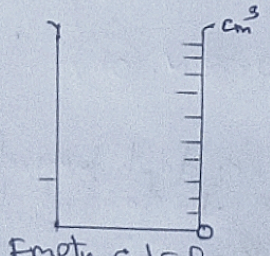
i	ii	iii	iv	v
H	F	C	E	A

01@ = 5marks

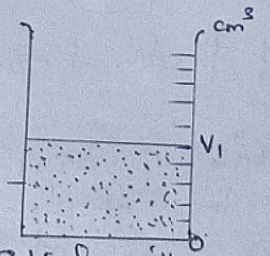
3@ Data
Least count = 0.01mm
Main scale = 1.6cm
= 16mm
Circular scale = 46 divisions (01)
= 46 × 0.01mm
= 0.46mm (01)
Total reading = main scale + circular scale
= 16mm + 0.46mm (02)
= 16.46mm
∴ Diameter of a ball bearing is 16.46mm. (01)

(b)(i) @ Number part, it tells about the magnitude of what to be measured.
(b) unit part, it tells about what is to be measured.

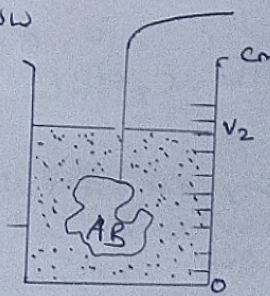
(ii) Consider the measuring cylinders below



Empty cylinder



Cylinder with liquid
01/2



Cylinder with liquid and irregular object
01/2

The volume of irregular object AB is determined by following the procedures below

i) Fill an empty measuring cylinder with a liquid to a certain level (V_1)

ii) Carefully measure the initial volume of liquid and name it as V_1

(iii) Gently Lower the irregular object into a liquid as shown in figure above

(iv) Read the final volume of liquid and name it V_2

(v) Find the difference in volume

difference = $V_2 - V_1$ = volume of an irregular object. (01½) = 02½

3(b) (iii) Because it contains three beams (01½)
100g graduations, 10g graduations, 1g graduation

4(a) (i) Data:
frictional force = 200N
John must apply a force of Magnitude greater or equal to the frictional force, therefore he must apply a force of 200N (02)

(ii) Mass = 100kg
Required, Coefficient of frictional force (μ)
from:

$\mu = \frac{f_r}{R}$ but $R = W = mg$.
$= 100kg \times 10N/kg$ (03)
$= 1000N$

$\mu = 200N / 1000N$ (02)
$= 0.2$

∴ The Coefficient of frictional force is 0.2N. (01)

(b) Cars are fixed with air-bags on their dash board in order to reduce the effects of force towards the driver when an accident occurs (02)

(i) (a) Density is the Mass of an object divide by volume of an object while Relative density is the ratio of density of object to density of water (02)

⇒ Density has SI-unit while relative density has no SI-unit.

(b) Mass is measured in Kg while Weight is measured in Newton. (Any difference) (02)

(ii) The density of water is 1000 kg/m³ this means that a capacity of 1 m³ can hold an amount of 1000kg. Therefore for every 1 m³ there are 1000kg of a substance (water). (01)

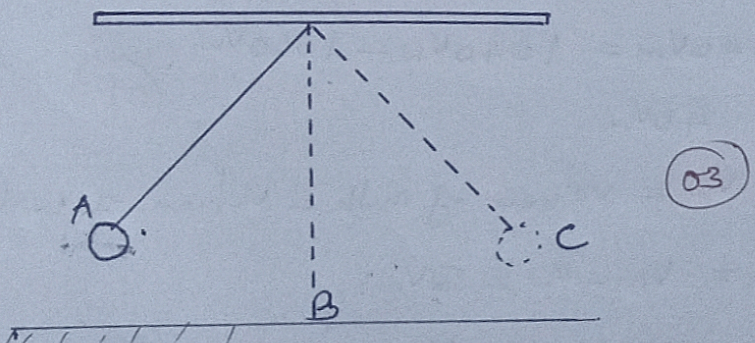
5(b) Applications of density

(i) Used in identification of gemstones eg: diamond, gold.
ii) Used by geologists to determine Mineral Content
iii) Used in selection of building materials
iv) Used during designing of swimming and diving equipment
v) Used in designing Marine Vehicles. 01 mark @ = 05

6@ (i) Transducer (01)

(ii) Kinetic energy is Converted to → Heat energy
→ Light energy 01@ = 03
→ Sound energy

6(b) A diagram of simple pendulum.



(03)

At point A there is maximum potential energy
At point B there is maximum kinetic energy
At point C there is maximum potential energy. (03)
At any point along AB & BC, the energy is Conserved, P.e + K.e = Constant.

Q(a) Data.

Density of milk = 1080 Kg/m³.

Density of diluted milk = 1040 Kg/m³

Required, Percentage of water in milk.

Mass of milk = M_m, volume of milk = V_m

Mass of water = M_w, volume of water = V_w

Density of mixture = $\frac{M_w + M_m}{V_w + V_m} = 1040 \text{ Kg/m}^3$

$$\frac{M_w + M_m}{V_w + V_m} = 1040 \longrightarrow (i) \quad (01)$$

but

$$\frac{M_m}{V_m} = 1080 \quad (01)$$

$$M_m = 1080 V_m \longrightarrow (ii)$$

Subst. (ii) into (i)

$$\frac{1080 V_m + M_w}{V_m + V_w} = 1040 \quad (01)$$

$$1080 V_m + M_w = 1040 V_m + 1040 V_w$$

$$\text{but } M_w = 1000 V_w \quad (01)$$

$$1080 V_m + 1000 V_w = 1040 V_m + 1040 V_w$$

$$1080 V_m - 1040 V_m = 1040 V_w - 1000 V_w$$

$$40 V_m = 40 V_w \quad (01)$$

$V_m = V_w$ = volume of milk = volume of water.

total volume = $V_m + V_w = 2V_m$.

Percentage of milk = $\frac{V_m}{2V_m} \times 100\%$, $\frac{V_w}{2V_w} \times 100\%$

$= \frac{1}{2} \times 100\%$, $\frac{1}{2} \times 100\%$ (01)

$= 50\%$, 50%.

∴ The volume percentage of water is 50%.

(i) A hollow ball has less density than solid ball
- A hollow ball has less weight than the upthrust compared to the weight of a solid ball. (02)

(ii) Any object when it is partially or totally immersed in a fluid it experiences an upthrust which is equal to the weight of the fluid displaced by the object. (02)

8@ Reasons for size of errors
(i) instrument fault during manufacture.
(ii) Damage during use
iii) Poor storage of an instrument. (04)
iv) Human factors (Parallax error, zero error, instrument error).

(b) Data:
Length = 4.00 cm
Width = 2.5 cm
Height = 1.00 cm

(i) The instrument used is meter rule (01)

(ii) volume = L x W x h.
= 4.00 cm x 2.5 cm x 1.00 cm
= 10 cm^3
volume is 10 cm^3 (01)

(iii) Relative density of
Density of rectangular block = $\frac{\text{Mass of block}}{\text{volume of block}}$ (01)

$\rho_b = \frac{300g}{10cm^3}$
$= 30g/cm^3$ (01)

Relative density = $\frac{\rho_b}{\rho_w}$
$= \frac{30g/cm^3}{1g/cm^3}$.
$= 30$ (02)

∴ Relative density of rectangular block is 30

9@ A good laboratory must have the following features.

(i) Physics Laboratory should be well ventilated and its doors should be open outwards.
(ii) Fire extinguisher should be fitted in accessible positions.
(iii) Laboratory floors should not be polished.
(iv) First aid kit should be present.
(v) Cabinets and drawers should be included during designing Laboratory room. (05)

9(b) (i) First Aid kit is a small box containing items that are used to give help to a sick or injured person while First Aid - is the first and immediate assistance or care given to a sick or injured person before getting a professional medical help. (05) (03)

(ii) → saves life
→ Prevent victim's condition from becoming worse
→ It promote hope and encouragement to a victim
→ It help to reduce pain and suffering
→ It prevents infections. (05) (04) (00½) @ = 2mark

10@ Data:

$M_e = 40g$
$M_e + M_s = 90g$
$M_s + M_{(ethanol)} = 120g$
$90g - 40g =$ Mass of sand
$M_s = 50g$ (03)
$M_{(ethanol)} = 120g - 50g$
$= 70g.$

(i) Mass of ethanol $= 70g = 0.07kg.$
(ii) Density of bottle $=$ Mass of bottle / Volume of bottle (01)
$= 40g / 600cm^3$
$= 0.067g/cm^3$
Density of bottle $= 0.067g/cm^3$ (01)

(iii) Relative density of an ethanol

Density of ethanol = $\frac{\text{Mass of ethanol}}{\text{volume of bottle}}$ (01)

$= 709 / 800\,cm^3$

$= 0.12\,g/cm^3$

Relative density = $\frac{\text{Density of ethanol}}{\text{Density of water}}$ (01)

$= \frac{0.12\,g/cm^3}{4\,g/cm^3}$

$= 0.12$

∴ Relative density of ethanol is 0.12. (01)

(iv) Comparison of densities of sand and ethanol.

density of sand = $\frac{\text{Mass of sand}}{\text{volume of bottle.}}$

$= \frac{50g}{400cm^3}$

$= 0.08\,g/cm^3$ (01)

Density of ethanol $= 0.12\,g/cm^3 > 0.08\,g/cm^3$

∴ Ethanol is more dense than sand. (01)

A2(b) Data:

Density of sea water (ρ_s) = $1100\,kg/m^3$

Volume of ship in sea water = V_s

Density of river water = $1000\,kg/m^3 = \rho_R$

volume of ship in river = V_r

Mass of ship is the same everywhere ($M_s = M_R$)

$M_s = \rho_s V_s$

$M_R = \rho_R V_R$

But $M_s = M_R$ (01)

$\rho_s V_s = \rho_R V_R$

Required: $\rho_s : \rho_R$, V_s / V_R :

$\frac{\rho_s}{\rho_R} = \frac{V_R}{V_s}$ (01)

Now =

$\frac{V_s}{V_R} = \frac{\rho_R}{\rho_s}$ (01)

$= \frac{1000 \text{ kg/m}^3}{1100 \text{ kg/m}^3}$

$= 10/11$ (01)

$V_s : V_R = 10 : 11$ (01)